

Tayyibah Khanam

Machine Learning Engineer
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EDUCATION

Degree	University	GPA	Year
Master of Science, Data Science	New York University, NY	3.73/4	2024
Bachelor of Technology, Electrical Engineering	Aligarh Muslim University, India	9.07/10	2022

SKILLS

- Languages/Tools: Python, SQL, AWS, Git, Latex, Linux, Airflow, W&B, DVC, Langchain, Langgraph
- Libraries: Pandas, NumPy, Matplotlib, Tensorflow, Pytorch, Scikit-Learn, Scipy, OpenCV, Plotly, Spacy, HuggingFace

EXPERIENCE

- **Kensho Technologies, S&P Global** Aug 2024 - Present
Machine Learning Engineer New York, NY
 - Designed and shipped the first statistical evaluation framework for RAG agents over unstructured financial data, now adopted as the company-wide standard for rigorous, reproducible agent comparison at scale.
 - Led a cross-team evaluation benchmarking the company’s document extraction-to-retrieval pipeline against open-source alternatives, providing quantitative evidence of superior performance that informed product strategy and leadership decisions.
 - Built and maintained production ML pipelines for document understanding models, including automated evaluation infrastructure, regression testing, and model upgrades — ensuring reliability and continuous improvement across the company’s core document extraction systems.
 - Owned an LLM-powered financial research assistant serving analysts end-to-end — optimizing retrieval workflows, prompt orchestration, and the LLM execution stack to improve citation accuracy and response quality across thousands of financial documents.
- **Columbia University** Sep 2023 - May 2024
Master’s Capstone New York, NY
 - Engineered multi-modal pipeline and fine-tuned LLMs, achieving 6.5% improvement in information extraction from long legal documents by merging Q&A data with HTML annotations, leveraging transformer models and HuggingFace.
 - Developed web application tools for efficient HTML annotations and NLP model interpretability, bolstering feedback mechanisms.
- **FOX Lab, NYU** Jan 2023 - Nov 2023
Graduate Research Assistant New York, NY
 - Developed a Python-based experimental pipeline analyzing features from time-series data of 260 songs & 640+ participants. Identified distinct listener subgroups through clustering & statistical modeling based on neural response patterns.
 - Achieved 71% accuracy in predicting user music preferences by ranking the top 5 influential song features, contributing original insights to music recommendation & generation research.
- **DataKind (Non-Profit)** Jan 2024 - May 2024
Data Science Intern Remote
 - Designed and built an end-to-end automated data pipeline for a social impact non-profit, processing 3,600+ event registrations and unstructured attendee metadata – reducing manual analysis time by 90% and directly informing organizational outreach strategy.
 - Led front-end development of a community web platform integrating feedback from social impact professionals, improving resource accessibility and stakeholder engagement for DataKind’s mission-driven initiatives.

RESEARCH

- [PubTables-v2: A Large-Scale Dataset for Full-Page and Multi-Page Table Extraction](#) | Submitted to ECCV 2026. B. Smock, V. Faucon-Morin, M. Sokolov, L. Liang, **T. Khanam**, A. Ramesh, M. Courtland.
- [Lightweight Deep Learning Model for Weed Detection for IoT Devices](#) | IEEE. **T. Khanam** *et al.*.
- [Efficient Machine Learning for Malnutrition Prediction Among Under-Five Children in India](#) | IEEE Xplore. **T. Khanam** *et al.*.
- [Experimentally Proven Bilateral Blur for Optimal Convergence](#) | Springer. D. Mishra, **T. Khanam**, I. Kaushik.
- [Optimizing Electric Vehicle Charging with Charging Data Analytics](#) | Springer. **T. Khanam** *et al.*.
- [Big Data Applications in Smart & Sustainable Energy Systems: Review & Case Study](#) | Springer. **T. Khanam** *et al.*.
- [Neuroscience 2023 Research Poster – Music Perception & Neural Response Modeling](#) | Society for Neuroscience Annual Meeting

HONORS & AWARDS

- **2nd Place, RutHai x Hearst Lab Hackathon**, New York, NY *Apr 2026*
- **Ranked 20th Globally**, EY Open Science Data Challenge – Fire Mapping from Satellite Imagery

MENTORSHIP & LEADERSHIP

- **GSTEM NYU – Mentorship & Teaching** *2024 - 2025*
Research Mentor & Teaching Assistant New York, NY
 - Mentored high school girls in hands-on STEM research projects as part of the GSTEM NYU Summer Program, guiding participants through research methodology, data analysis, and scientific communication.
 - Served as Teaching Assistant for the GSTEM Spring Data Science course, supporting curriculum delivery and helping students develop foundational skills in Python, data analysis, and machine learning.